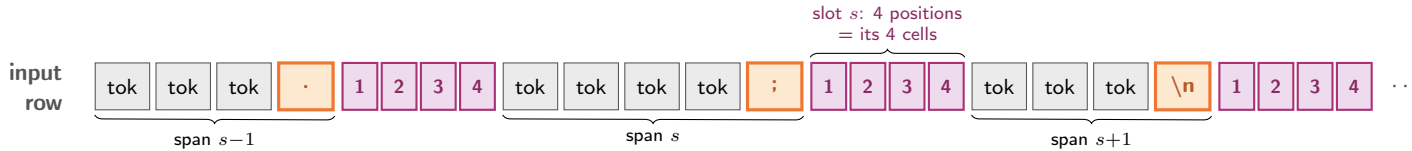


LXTUL: the latent-selected slot loop

primary candidate, morph/configs/lxtul.yaml (locked 2026-10-04)

one row = 1024 tokens; after every span the packer inserts 4 slot positions (≤ 64 slots, row length $1024 + 4 \cdot 64 = 1280$)
orange = boundary (., !?, newline, dashes; no comma) · span length 4–32 · every slot position's input = $E_{\text{slot}} + W_{\text{sent}} \text{embed}(t_{\text{last}})$



STRICT GEOMETRY: the slot loop is the only path from one span to a later span

stage	a token reads	a slot position / cell reads
prelude, 4 blocks	its own span's tokens only	its own span's tokens and its own earlier slot positions
slot loop, 6 blocks $\times T$	(tokens are not in the loop)	all 4 cells of its own slot + every cell of every earlier slot
coda, 4 blocks	its own span's tokens + the prefix positions of every earlier slot	itself only (it carries what the loop wrote, nothing else)

conv, value shift and retention carry reset at every segment; the coda's per-layer injections at slot positions are zeroed

PRELUDE (4 blocks): every position of the row, once

slot positions

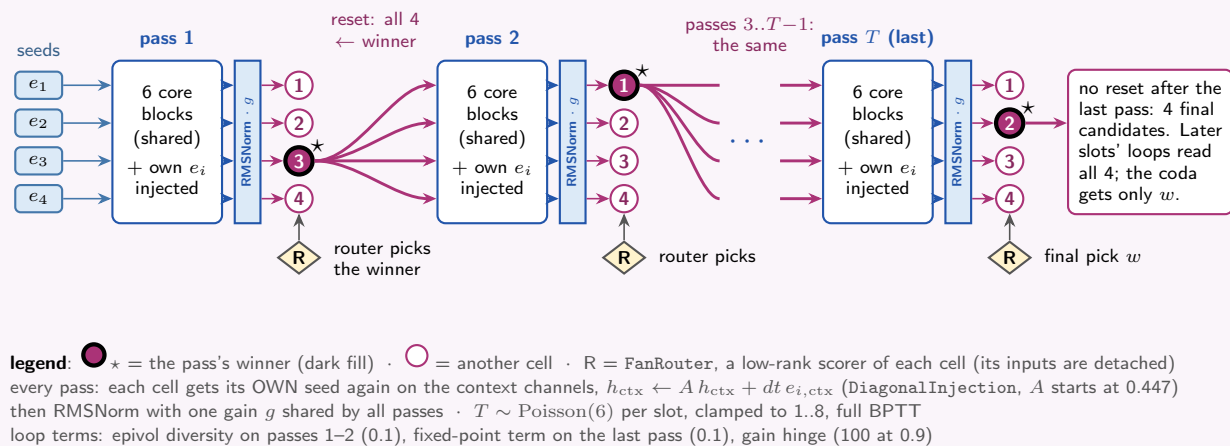
SEED the 4 cells of slot s (TULSlotRegister), $i = 1..4$:

$$e_i = \hat{x}_{s,i} + W_o \text{Attn}(Q_i; x_{j \in \text{span } s}) + P_{\text{cell}}[i]$$

$\hat{x}_{s,i}$ = normed prelude state at slot position i · Q_i = cell i 's own learned query, one head, keys and values = span s 's prelude token states
 W_o and P_{cell} start at zero · cell carrier = Hyper-Connection, 4 streams $\times d$ 1024 · $h_0 = e_i$

$e_1..e_4$ per slot

THINK: every slot's 4 cells loop together; after each pass the router picks one cell and all 4 restart from it

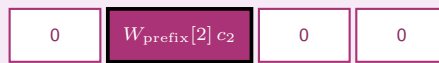


winner w of every slot

WRITE: the winner alone

slot position $k \leftarrow W_{\text{prefix}}[k] c_k$ for $k = w$; the 3 other positions are EXACTLY 0
so the coda reads ONE live vector per earlier span

slot s 's 4 positions ($w = 2$ here)



CODA (4 blocks): every position, once $\rightarrow$ LM head $\rightarrow$ next-token CE

a token reads its own span + the prefix positions of every earlier slot · token-state dropout 0.15
no CE at slot positions; the boundary token keeps its ordinary label

tokens: prelude state straight to the coda, never in the loop

TRAIN ONLY: the router's teacher

- an EMA twin of the prelude (momentum 0.996, no gradient) reads span $s+1$: $z_s = \text{LayerNorm}(\text{mean of its token states})$
- head g (LayerNorm, MLP) reads the cells DETACHED; teacher pick: $\arg \min_i \|g(c_i) - z_s\|^2$
- router: cross-entropy onto the teacher pick, at every pass
- exit loss: relaxed winner-take-all on g (weight 1) trains g only, so no latent gradient reaches the loop; a std floor on the online prelude's pooled span states (0.2)

The forward follows the router, at train and at eval. z is a label only: nothing built from it enters the loop or the coda.

TRAIN ONLY: span decoder

2 layers, weight 1: teacher-forced decode of the NEXT span (span $s+1$). It reads the uniform MEAN of the 4 final candidates, not the winner.

GENERATION

a new token costs its prelude pass and one coda pass, no core pass · an emitted boundary inserts 4 slot positions and costs ONE think for that new slot (T passes $\times$ 4 cells)
today only the eager recompute generator (tul_generate.py) runs LXTUL: the KV-cached generators refuse $\text{fan_k} > 0$, so decode speed is not yet measured

MEASURED (480 sweep rows, paired CIs)

K1–K6 (what loop depth earns): +0.0215 / +0.0172 at 5k (two seeds), +0.0236 at 10k
CE gap to a plain model at the same step: +0.272 (5k), +0.356 (10k)
65% of the 10k gap is bigram repeats 33–256 tokens back (15% of tokens)

lab/experiments/successes/2026-10-03-rank-only-cell-norm-seed-twin.md,
2026-10-04-rank-only-cell-norm-10k.md, 2026-10-04-exact-recall-gap-probe.md